

The screenshot displays the Amazon Athena Query Editor interface. The top navigation bar shows the AWS logo, a search bar, and the account ID: 3397-1286-2909. The main content area is divided into a left sidebar and a main workspace.

Left Sidebar:

- Data source:** AwsDataCatalog
- Database:** taxidata
- Tables and views:** A search bar with the text "Filter tables and views". Below it, a list of tables is shown: "yellow" (1 table).

Main Workspace:

- SQL Editor:** Contains the query: `SELECT * FROM "taxidata"."yellow" limit 10;`
- Query Execution:** Buttons for "Run again", "Explain", "Cancel", "Clear", and "Create". A status bar indicates "Completed" with a green checkmark.
- Query Stats:** Time in queue: 96 ms, Run time: 1.372 sec, Data scanned: 687.00 KB.
- Results (10):** A table showing 10 rows of data. The table has columns: #, vendor, pickup, dropoff, count, distance, ratecode, storeflag, and pulocid.

Results Table:

#	vendor	pickup	dropoff	count	distance	ratecode	storeflag	pulocid
1	1	2017-10-02 13:18:58.000	2017-10-02 13:26:41.000	2	1	1	N	231
2	1	2017-10-02 13:45:29.000	2017-10-02 14:07:11.000	1	2	1	N	249
3	1	2017-10-02 13:01:37.000	2017-10-02 13:08:33.000	1	0	1	N	237
4	1	2017-10-02 13:14:45.000	2017-10-02 13:28:22.000	1	1	1	N	237
5	1	2017-10-02 13:36:41.000	2017-10-02 13:53:03.000	1	2	1	N	140
6	1	2017-10-02 13:31:57.000	2017-10-02 13:51:42.000	2	2	1	N	239
7	1	2017-10-02 13:08:10.000	2017-10-02 13:14:17.000	1	0	1	N	236
8	1	2017-10-02 13:18:19.000	2017-10-02 13:21:43.000	1	0	1	N	237
9	1	2017-10-02 13:24:10.000	2017-10-02 13:29:33.000	1	0	1	N	237
10	1	2017-10-02 13:37:20.000	2017-10-02 14:00:06.000	2	2	1	N	236

Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/b645d7a4-7a5a-4b97-b9f3-404bd2e1f1cb

Account ID: 3397-1286-2909
voclabs/user4454514-bssie23058@itu.edu.pk

Amazon Athena > Query editor

views (0)

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 68 ms Run time: 11.289 sec Data scanned: 9.32 GB

Results (2,383,452) Copy Download results CSV

Search rows

#	Number of trips	Total fares	Trip date
1	1	10	2017-01-15 03:23:52.000
2	4	84	2017-01-15 03:24:47.000
3	1	7	2017-01-15 03:24:51.000
4	2	26	2017-01-15 03:26:20.000
5	3	25	2017-01-15 03:28:48.000
6	3	78	2017-01-15 03:29:29.000
7	3	38	2017-01-15 03:30:36.000
8	2	45	2017-01-15 03:30:43.000

CloudShell Feedback

Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/b645d7a4-7a5a-4b97-b9f3-404bd2e1f1cb

Account ID: 3397-1286-2909
voclabs/user4454514-bssie23058@itu.edu.pk

Amazon Athena > Query editor

#	Number of trips	Total fares	Trip date
10	2	33	2017-01-15 03:31:16.000
11	5	65	2017-01-15 03:33:04.000
12	2	28	2017-01-15 03:38:34.000
13	3	27	2017-01-15 03:40:36.000
14	3	47	2017-01-15 03:41:24.000
15	6	130	2017-01-15 03:41:44.000
16	2	21	2017-01-15 03:42:24.000
17	3	43	2017-01-15 03:42:45.000
18	3	83	2017-01-15 03:42:57.000
19	3	58	2017-01-15 03:48:34.000
20	2	11	2017-01-15 03:50:55.000
21	2	45	2017-01-15 03:51:43.000
22	1	21	2017-01-15 03:53:10.000
23	3	67	2017-01-15 03:53:49.000
24	2	44	2017-01-15 03:55:45.000

Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/4f6b9144-083a-4dc3-8e8c-05013fc5c95e

Account ID: 3197-1286-2909
voclabs/user4454514-bssie23058@itu.edu.pk

Amazon Athena > Query editor

Completed Time in queue: 68 ms Run time: 6.366 sec Data scanned: 815.32 MB

Results (2,383,451) [Copy](#) [Download results CSV](#)

Search rows

#	Number of trips	Total fares	Trip date
1	2	26	2017-01-30 07:56:00.000
2	6	71	2017-01-30 07:56:02.000
3	7	87	2017-01-30 07:56:36.000
4	4	43	2017-01-30 07:56:40.000
5	3	31	2017-01-30 07:56:48.000
6	3	25	2017-01-30 07:57:28.000
7	1	6	2017-01-30 07:58:08.000
8	5	64	2017-01-30 07:58:37.000
9	8	124	2017-01-30 07:59:04.000
10	4	31	2017-01-30 07:59:09.000
11	3	32	2017-01-30 07:59:58.000

CloudShell Feedback

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Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/4f6b9144-083a-4dc3-8e8c-05013fc5c95e

Account ID: 3197-1286-2909
voclabs/user4454514-bssie23058@itu.edu.pk

Amazon Athena > Query editor

#	Number of trips	Total fares	Trip date
13	9	144	2017-01-30 08:00:26.000
14	3	30	2017-01-30 08:00:51.000
15	7	130	2017-01-30 08:01:19.000
16	2	19	2017-01-30 08:01:39.000
17	3	48	2017-01-30 08:02:03.000
18	4	40	2017-01-30 08:02:07.000
19	6	60	2017-01-30 08:02:25.000
20	9	196	2017-01-30 08:02:56.000
21	2	24	2017-01-30 08:02:59.000
22	3	39	2017-01-30 08:03:00.000
23	3	30	2017-01-30 08:03:01.000
24	5	66	2017-01-30 08:03:21.000
25	5	41	2017-01-30 08:03:48.000
26	6	87	2017-01-30 08:05:38.000
27	3	79	2017-01-30 08:05:41.000

CloudShell Feedback

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Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/29715b59-8179-4e9f-a673-27a73c513f71

Account ID: 3397-1286-2909
voclabs/user4454514-bssie23058@nu.edu.pk

Amazon Athena > Query editor

Data source: AwsDataCatalog
Database: taxidata
Tables and views: creditcard, jan, yellow
Views (0)

SQL
Ln 2, Col 40

```
1 SELECT sum (total), paytype FROM yellow  
2 WHERE paytype = '1' GROUP BY paytype;
```

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 102 ms Run time: 4.021 sec Data scanned: 9.32 GB

Results (1) Copy Download results CSV

Search rows

#	_col0	paytype
1	1351232654	1

Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/9bd7890f-0004-4bde-aa92-67dec257fed1

Amazon Athena > Query editor

Editor Recent queries Saved queries Query settings Workgroup primary

Athena now supports typeahead code suggestions to speed up SQL query development
Typeahead suggestions are turned on by default. You can change this setting in query editor preferences. Edit preferences

Data

Data source
AwsDataCatalog

Database
taxidata

Tables and views Create Filter tables and views

Tables (3)
creditcard
jan
yellow

Views (0)

```
1 SELECT sum (total), paytype FROM creditcard  
2 WHERE paytype = '1' GROUP BY paytype;
```

SQL Ln 2, Col 40

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

CloudShell Feedback

Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/9bd7890f-0004-4bde-aa92-67dec257fed1

Amazon Athena > Query editor

Tables and views Create Filter tables and views

Tables (3)
creditcard
jan
yellow

Views (0)

SQL Ln 2, Col 40

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 113 ms Run time: 1.789 sec Data scanned: 71.81 MB

Results (1) Copy Download results CSV

Search rows

#	_col0	paytype
1	1351232654	1

Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/athena/home?region=us-east-1#/query-editor/history/7dee355b-49a8-4b88-95d7-dac4a23d421

Amazon Athena > Query editor

Tables and views

Filter tables and views

Tables (3)

- creditcard
- jan
- yellow

Views (3)

- cashtrips
- cctrips
- comparepay

SQL Ln 1, Col 1

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 104 ms Run time: 6.523 sec Data scanned: 18.64 GB

Results (2)

Search rows

#	cctotal	cashtotal
1	460097126	199181978
2	584502884	250849783

CloudShell Feedback

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Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/cloud9/ide/e836d1762201498f9c079cd17129928?region=us-east-1

File Edit Find View Go Run Tools Window Support

athenaquery.cf.yml

```

1 AWSTemplateFormatVersion: 2010-09-09
2 Resources:
3   AthenaNamedQuery:
4     Type: AWS::Athena::NamedQuery
5     Properties:
6       Database: "taxidata"
7       Description: "A query that selects all fares over $100.00 (US)"
8       Name: "FaresOver100DollarsUS"
9       QueryString:
10        SELECT distance, paytype, fare, tip, tolls, surcharge, total
11        FROM yellow
12        WHERE total >= 100.0
13        ORDER BY total DESC
14 Outputs:
15   AthenaNamedQuery:
16     Value: !Ref AthenaNamedQuery

```

bash - "ip-10-16-10-175.ec2" Immediate

```

voclabs:~/environment $ aws cloudformation validate-template --template-body file://athenaquery.cf.yml
{
  "Parameters": []
}
voclabs:~/environment $ aws cloudformation create-stack --stack-name athenaquery --template-body file://athenaquery.cf.yml
{"StackId": "arn:aws:cloudformation:us-east-1:339712862909:stack/athenaquery/e8b69b3b-d266-11f0-a2d2-0afdf5e2bde6"}
voclabs:~/environment $ aws athena list-named-queries

```

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Query editor | Athena | us-east-1

us-east-1.console.aws.amazon.com/cloud9/ide/e836d1762201498f9c079cd17129928?region=us-east-1

File Edit Find View Go Run Tools Window Support

athenaquery.cf.yml

```

1 AWSTemplateFormatVersion: 2010-09-09
2 Resources:
3   AthenaNamedQuery:
4     Type: AWS::Athena::NamedQuery
5     Properties:
6       Database: "taxidata"
7       Description: "A query that selects all fares over $100.00 (US)"
8       Name: "FaresOver100DollarsUS"
9       QueryString:
10        SELECT distance, paytype, fare, tip, tolls, surcharge, total
11        FROM yellow
12        WHERE total >= 100.0
13        ORDER BY total DESC
14 Outputs:
15   AthenaNamedQuery:
16     Value: !Ref AthenaNamedQuery

```

bash - "ip-10-16-10-175.ec2" Immediate

```

voclabs:~/environment $ aws athena list-named-queries
{
  "NamedQueryIds": [
    "701398f2-7211-4b07-92fc-d1fe1184d96d"
  ]
}
voclabs:~/environment $ aws athena get-named-query --named-query-id 701398f2-7211-4b07-92fc-d1fe1184d96d
{
  "NamedQuery": {
    "Name": "FaresOver100DollarsUS",
    "Description": "A query that selects all fares over $100.00 (US)",
    "Database": "taxidata",
    "QueryString": "SELECT distance, paytype, fare, tip, tolls, surcharge, total FROM yellow WHERE total >= 100.0 ORDER BY total DESC\n",
    "NamedQueryId": "701398f2-7211-4b07-92fc-d1fe1184d96d",
    "WorkGroup": "primary"
  }
}
voclabs:~/environment $

```

AWS profile: default

```
1 AKSTemplateFormatVersion: 2018-09-09
2 Resources:
3   AthenaNamedQuery:
4     Type: AWS::Athena::NamedQuery
5     Properties:
6       Database: "taxidata"
7       Description: "A query that selects all fares over $100.00 (US)"
8       Name: "FaresOver100DollarsUS"
9       QueryString: ?
10      SELECT distance, paytype, fare, tip, tolls, surcharge, total
11      FROM yellow
12      WHERE total >= 100.0
13      ORDER BY total DESC
14     Outputs:
15       AthenaNamedQuery:
16         Value: !Ref AthenaNamedQuery
```

```
bash - "ip-10-16-10-175.ec2" Immediate
{"NamedQueryId": [
  "NamedQueryId": [
    "701398f2-7211-4b07-92fc-dfe1184d96d"
  ]
}]
voclabs:~/environment $ aws athena get-named-query --named-query-id 701398f2-7211-4b07-92fc-dfe1184d96d
{
  "NamedQuery": {
    "Name": "FaresOver100DollarsUS",
    "Description": "A query that selects all fares over $100.00 (US)",
    "Database": "taxidata",
    "QueryString": "SELECT distance, paytype, fare, tip, tolls, surcharge, total FROM yellow WHERE total >= 100.0 ORDER BY total DESC\n",
    "NamedQueryId": "701398f2-7211-4b07-92fc-dfe1184d96d",
    "WorkGroup": "primary"
  }
}
voclabs:~/environment $ aws athena get-named-query --named-query-id 701398f2-7211-4b07-92fc-dfe1184d96d
voclabs:~/environment $ echo $NQ
701398f2-7211-4b07-92fc-dfe1184d96d
voclabs:~/environment $
```

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DataScienceGroup Info

Summary

User group name: DataScienceGroup

Creation time: December 06, 2025, 10:11 (UTC+05:00)

ARN: arn:aws:iam::339712862909:group/DataScienceGroup

Permissions policies (1) Info

You can attach up to 10 managed policies.

Policy name	Type	Attached entities
Policy-For-Data-Scientists	Customer managed	1

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The screenshot shows the AWS Cloud9 IDE interface. The top browser tabs include 'Query editor | Athena | us-east-1', 'Stack c191063a492311213006', 'DataScienceGroup | IAM | Glob...', and 'Cloud9 Instance - AWS Cloud9'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/cloud9/ide/e836d1762201498f9c079cd171299287?region=us-east-1`.

The IDE has a menu bar (File, Edit, Find, View, Go, Run, Tools, Window, Support) and a toolbar. The left sidebar shows the file explorer with 'athenaquery.cf.yml' and 'README.md' under 'Cloud9 Instance -'. The main editor displays the 'athenaquery.cf.yml' file with the following content:

```
1 AKSTemplateFormatVersion: 2018-09-09
2 Resources:
3   AthenaNamedQuery:
4     Type: AWS::Athena::NamedQuery
5     Properties:
6       Database: "taxidata"
7       Description: "A query that selects all fares over $100.00 (US)"
8       Name: "FaresOver100DollarsUS"
9       QueryString: >
10         SELECT distance, paytype, fare, tip, tolls, surcharge, total
11         FROM yellow
12         WHERE total >= 100.0
13         ORDER BY total DESC
14     Outputs:
15       AthenaNamedQuery:
16         Value: !Ref AthenaNamedQuery
```

The bottom pane shows a terminal window with the following commands and output:

```
bash - "ip-10-16-10-175.ec2" Immediate
voclabs:~/environment $ echo $AQ
701398f2-7211-4b07-92fc-dfe1184d96d
voclabs:~/environment $ AK=
voclabs:~/environment $ clear
voclabs:~/environment $ AK=AKIAUGGDM5K6Y2GVBPJ
voclabs:~/environment $ SAK=equ8RCHL7yvpd1Qe82w2fAG3PjBld/3GzMLrep
voclabs:~/environment $ AWS_ACCESS_KEY_ID=$AK AWS_SECRET_ACCESS_KEY=$SAK aws athena get-named-query --named-query-id $AQ
{
  "NamedQuery": {
    "Name": "FaresOver100DollarsUS",
    "Description": "A query that selects all fares over $100.00 (US)",
    "Database": "taxidata",
    "QueryString": "SELECT distance, paytype, fare, tip, tolls, surcharge, total FROM yellow WHERE total >= 100.0 ORDER BY total DESC\n",
    "NamedQueryId": "701398f2-7211-4b07-92fc-dfe1184d96d",
    "WorkGroup": "primary"
  }
}
voclabs:~/environment $
```

The status bar at the bottom left indicates 'AWS profile: default'.

33 (Grades)

Lab: Querying Data by Using Athena

01:31

Submit Submission Report Grades

Important: Some of the checks made by the submission process in this lab will only give you credit if it has been at least 5 minutes since you completed the action. If you do not receive credit the first time you submit, you may need to wait a couple minutes and the submit again to receive credit for these items.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is recorded for this lab.

34. To find detailed feedback about your work, choose **Submission Report**.

Lab complete

Congratulations! You have completed the lab.

35. At the top of this page, choose **End Lab**, and then choose **Yes** to confirm that you want to end the lab.

A message panel indicates that the lab is terminating.

36. To close the panel, choose **Close** in the upper-right corner.

Task	Score
Total score	45/45
[Task 1a] Database created	5/5
[Task 1b] Database table created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5

Previous Next

Lab: Querying Data by Using Athena

01:30

Submit Submission Report Grades

Important: Some of the checks made by the submission process in this lab will only give you credit if it has been at least 5 minutes since you completed the action. If you do not receive credit the first time you submit, you may need to wait a couple minutes and the submit again to receive credit for these items.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is recorded for this lab.

34. To find detailed feedback about your work, choose **Submission Report**.

Lab complete

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Task	Score
[Task 1a] Database created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5
[Task 7] GetNamedQuery event recorded	5/5

Previous Next

34 (Submission report)

Lab: Querying Data by Using Athena

Submission Report

```
[Executed at: Fri Dec 5 21:55:39 PST 2025]

gradeFile = /mnt/vocwork2/grader/eee_g_2538066/asn4923111_2/asn4923112_1/tmp/temp_uf_12052025/.y0wtdK
reportFile = /mnt/vocwork2/grader/eee_g_2538066/asn4923111_2/asn4923112_1/tmp/temp_uf_12052025/.lNg9OR
/mnt/vocwork2/grader/eee_g_2538066/asn4923111_2/asn4923112_1/tmp/temp_uf_12052025/.y0wtdK
/usr/lib/python3.7/site-packages/boto3/compat.py:82: PythonDeprecationWarning: Boto3 will no longer support Py
warnings.warn(warning, PythonDeprecationWarning)
Started: 2025-12-05 21:55:13
region: us-east-1
profile: default

Evaluating [Task 1a] Database created
```

Congratulations! You have completed the lab.

35. At the top of this page, choose **End Lab**, and then choose **Yes** to confirm

A message panel indicates that the lab is terminating.

36. To close the panel, choose **Close** in the upper-right corner.

Lab: Querying Data by Using Athena

Submission Report

```
found matching database: taxidata

Task 1a Testing report - Success! An AWS Glue database named taxidata was found.

Evaluating [Task 1b] Database table created

found matching table: yellow

Task 1b Testing report - Success! An AWS Glue database table named yellow was found.

Evaluating [Task 1c] Database table previewed

found SELECT * FROM "taxidata"."yellow" limit 10 in query id: bcd2a81d-bbfd-4b76-8f68-e94d56d124c0
```

Congratulations! You have completed the lab.

35. At the top of this page, choose **End Lab**, and then choose **Yes** to confirm

A message panel indicates that the lab is terminating.

36. To close the panel, choose **Close** in the upper-right corner.

The screenshot displays the AWS Academy console interface. A 'Submission Report' modal is open, showing the following details:

- Submission Report**
- 2025-12-06T05:46:36Z
- ARQAU6GDW5K63WO32EOHE:user:4454514=bsse23058@itu.edu.pk
- CreateStack
- 4abc581e-e070-47cc-b3b8-e902c1add6a3
- ["eventVersion": "1.11", "userIdentity": {"type": "AssumedRole", "principalId": "ARQAU6GDW5K63WO32EOHE:user:4454514=bsse23058@itu.edu.pk"}]
- Task 5b Testing report - Success! A CloudFormation stack was created with the CreateStack command.
- Evaluating [Task 5c] CreateNamedQuery event recorded
- user:4454514=bsse23058@itu.edu.pk created a named query event with 10720b34-ae08-4850-bd0d-0ec7fe234235

The background interface shows the 'Lab: Querying Data by Using Athena' page. The sidebar menu includes Home, Discussions, Grades, Lucid (Whiteboard), Assignments, and Modules. The main content area displays a list of tasks, with the current task being 'Task 5b Testing report - Success! A CloudFormation stack was created with the CreateStack command.' The task status is 'Succeeded'.

Lab: Querying Data by Using Athena

awsacademy.instructure.com/courses/148677/modules/items/14403162

ADEv1EN-US-LTI13-148677 > Modules > Module 4 Design Principles and Patterns for Data Pipelines > Lab: Querying Data by Using Athena

Home
Discussions
Grades
Lucid (Whiteboard)
Assignments
Modules

Submission Report

found matching first column name with vendor. The columns are updated for the Glue table are updated.

Task 2 Testing report - Success! AWS Glue database table named jan created.

Evaluating [Task 4] Student queried using views

found Select * from cctrips in query id: 294aa4be-38a4-4f28-bc2b-8c00d24d5630

Task 4 Testing report - Success! A view was created in Athena.

Evaluating [Task 5a] Student tested the query in Athena

found SELECT distance, paytype, fare, tip, tolls, surcharge, total FROM yellow WHERE total >= 100.0 ORDER BY total

Task 5a Testing report - Success! You tested the query Mary gave you in Athena.

33. At the top of this page, choose **End Lab**, and then choose **Yes** to confirm

A message panel indicates that the lab is terminating.

36. To close the panel, choose **Close** in the upper-right corner.